



Panorama-Sampler



Alternative to AWR and ASH,
license-free for Oracle DB Standard
Edition



About Otto Group one.O

Fusion to one.O
2025

Parent company
Otto Group

Locations
Dresden, Hamburg, Altenkunstadt, Madrid,
Málaga, Valencia, Taipei, Hyderabad

Number of employees
around 1,000

Management Board
Katrin Behrens, Dr. Stefan Borsutzky

About me

Peter Ramm

Software architekt / Team lead at One.O in Dresden

More than 30 years of history in IT-projects

Main focus:

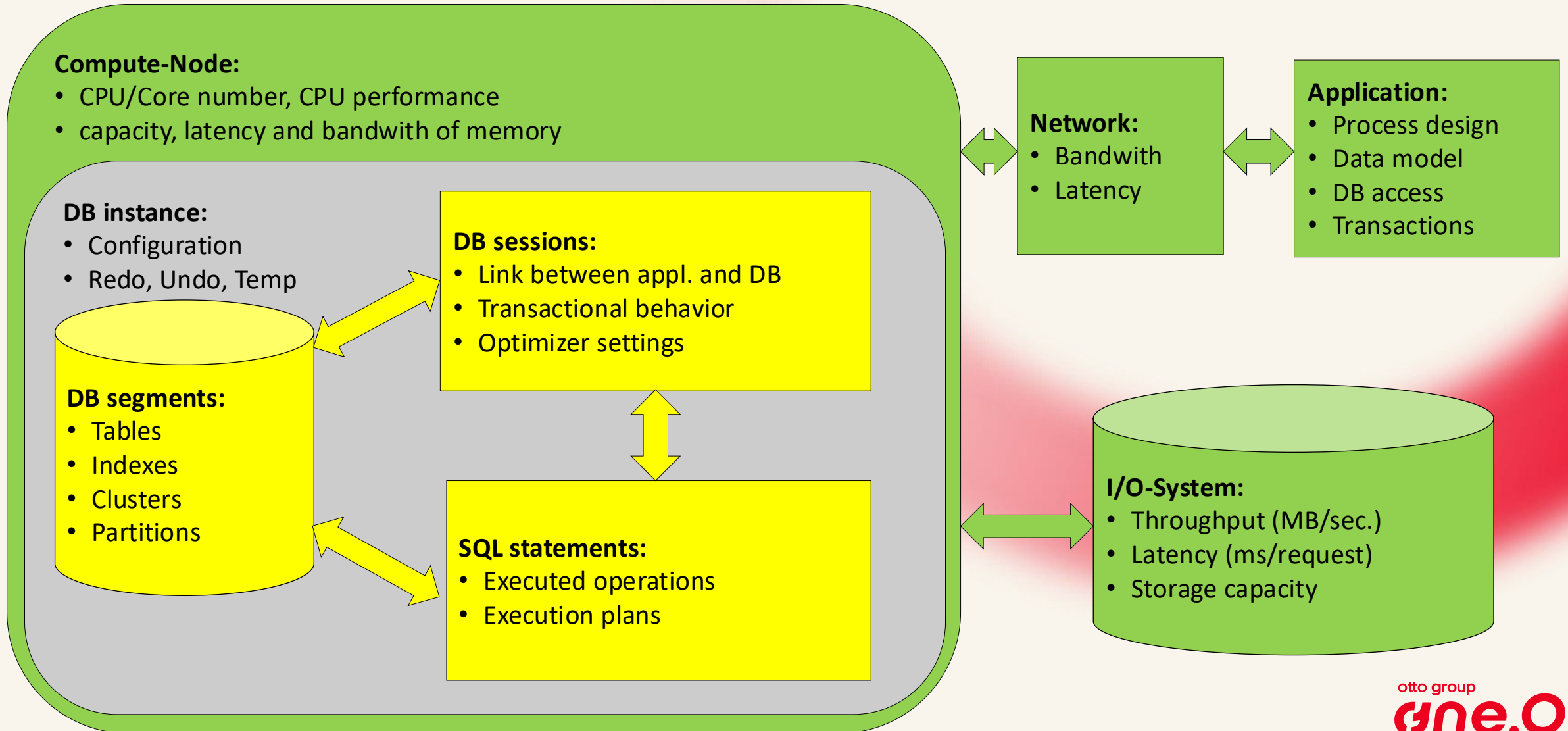
- Development of OLTP systems based on Oracle databases
- From architecture consulting up to trouble shooting
- Performance analysis and optimization of working systems



Oracle Ace Associate since 2024

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Factors influencing performance in DB use



Oracles solution for historization of metrics and statistics

- StatsPack: own rudimentary snapshots, available since release 8i
- Active Workload Repository (AWR), since release 10g
- Active Session History (ASH), since release 10g
- AWR and ASH are tailor-made for troubleshooting and forensics
- But require licensing of Enterprise Edition + Diagnostics Pack

Why not simply use the builtin function of Diagnostics Pack

Oracle Database pricing 2025

License	US\$ / processor	Support US\$ / year (22%)
Standard Edition 2	17,500	3,850
Enterprise Edition	47,500	10,450
Diagnostics Pack	7,500	1,650
Enterprise Edition + Diagnostics Pack	55,000	12,100

Solution approach for a workload repository similar to AWR

- Extend the existing performance analysis tool “Panorama” with own snapshots of metrics and statistics
- Records in tables that have the same structure as their AWR counterparts (DBA_HIST_xxx views) in rel. 19 or 23.
- Same name suffixes for tables (PANORAMA_xxx instead of DBA_HIST_xxx)
- The identical structures make it possible to:
 - Easy switching in Panorama between AWR/ASH as database or own recordings
 - All Panorama functions based on AWR/ASH are thus also available for SE and EE without Diagnostics Pack
 - Other analysis software can also be used without license violation by redirecting the access on AWR/ASH views to the Panorama replacement by synonyms.

Panorama for Oracle Databases

Tool for performance analysis of Oracle DB Free of charge (GPL3)

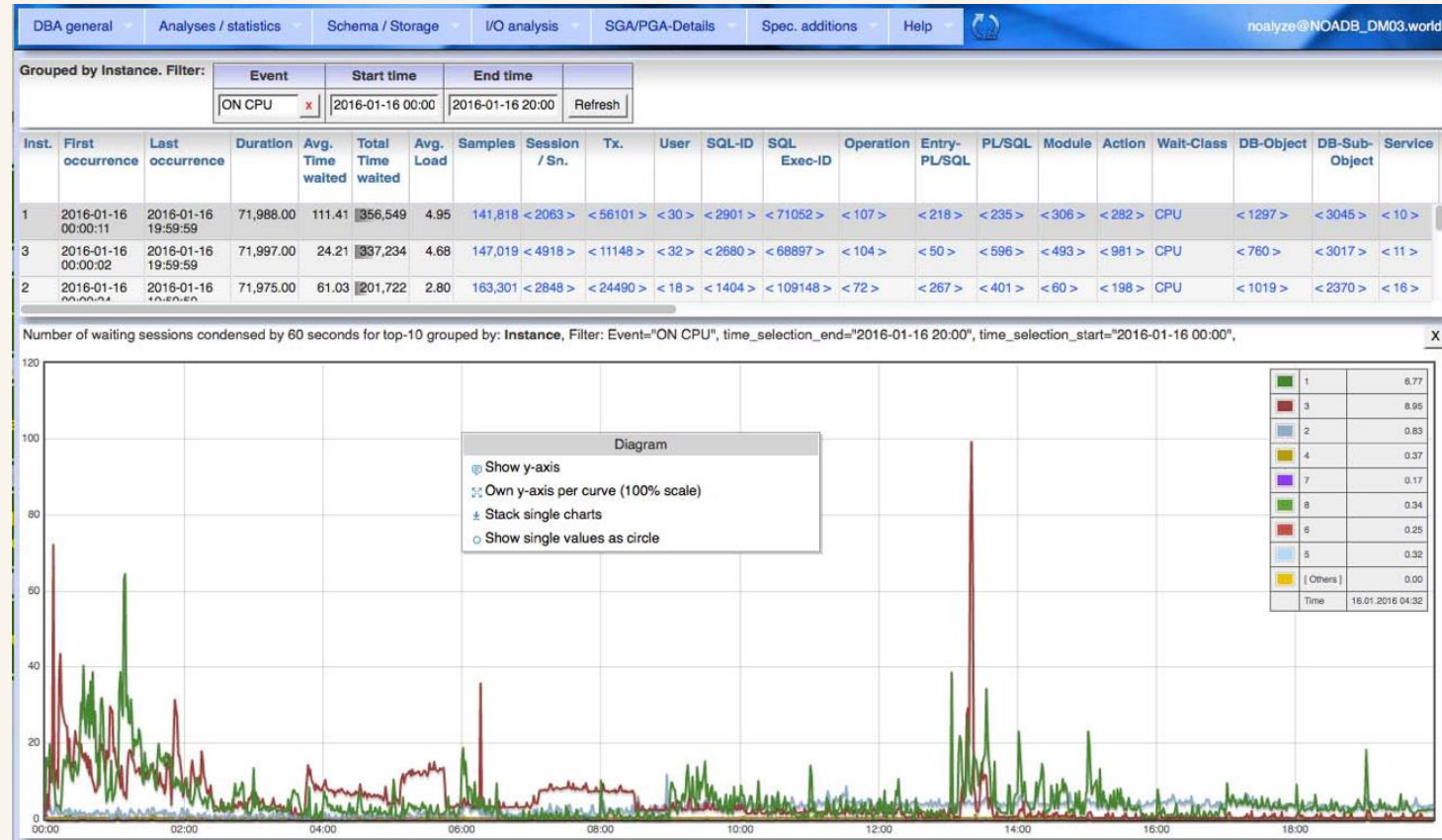
Based on dynamic performance views and
- AWR recordings
- or Panorama's own sampling
(for use with SE or without Diagnostics Pack)

Docker image or self-starting jar file

Panorama accesses your DB on a read-only basis and does not install any DB objects of its own. (except using Panorama-Sampler)

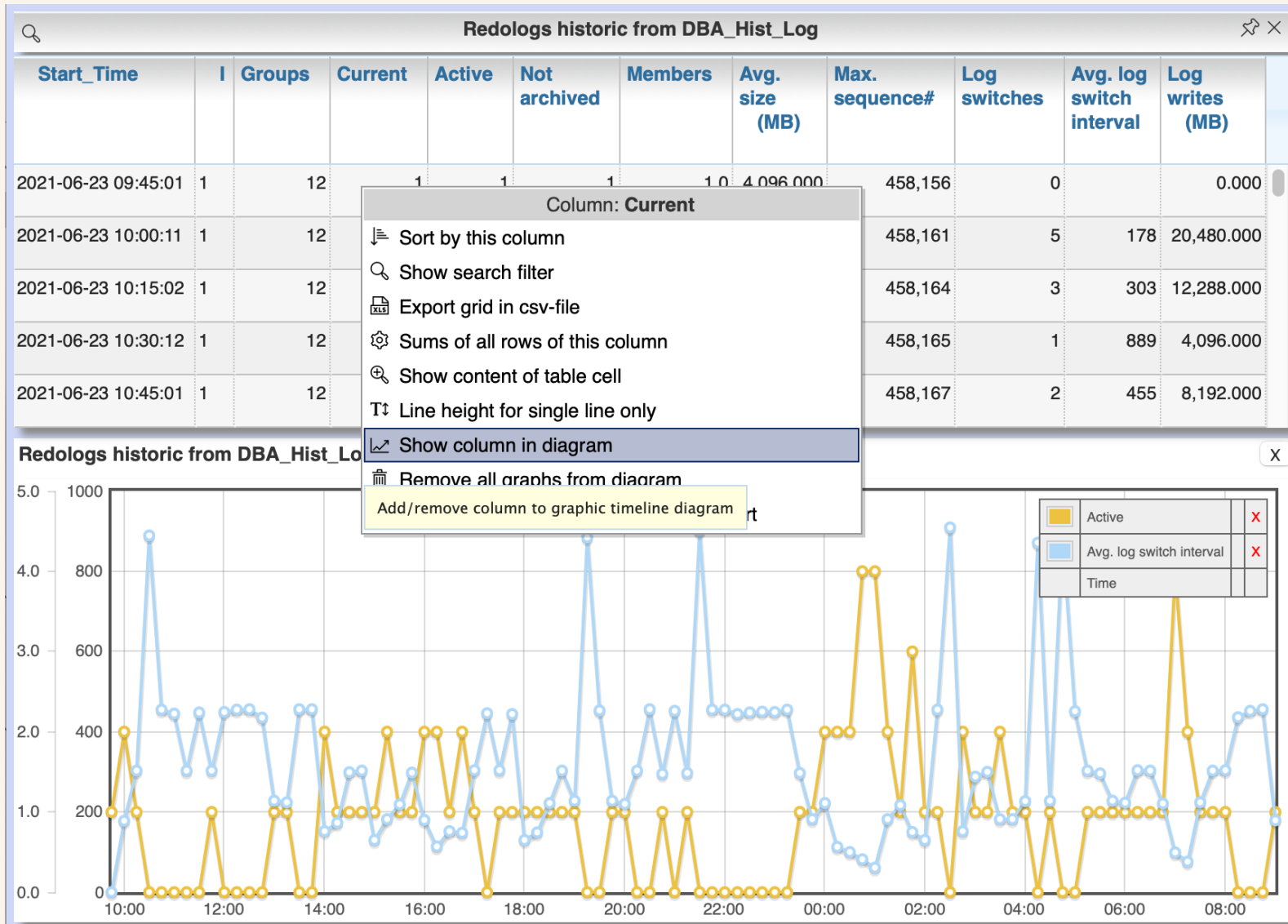
So you can test the functions without any risk.

Description of Panorama incl. download link :
Oracle performance analysis blog :
Various slides on the use of Panorama :



<https://rammpeter.github.io/panorama.html>
<https://rammpeter.blogspot.com>
<https://www.slideshare.net/PeterRamm1>

Panorama for Oracle: presentation formats



Workflow in browser page continuously downwards

Results usually as tables

Time-related values can generally also be displayed as graphs by showing/hiding columns of the tables

Visualization in graphs via context menu (right mouse button)

Config setting
PANORAMA_LOG_LEVEL=debug
shows executed SQLs on console

Technical implementation of the Panorama-Sampler

- Recording is done by PL/SQL code at the target DB, data is stored locally in that DB
- No network transfer of sampled data to the Panorama server
- ASH recording is done by a permanent background session
- A Panorama server instance can handle an unlimited number of target DBs
- Sampling is done independently in own thread per target DB
- The needed DB objects in target DB are created automatically at first use
- Supports all DB releases and editions starting with 11.2 up to 26ai

AWR views with replacement in Panorama-Sampler

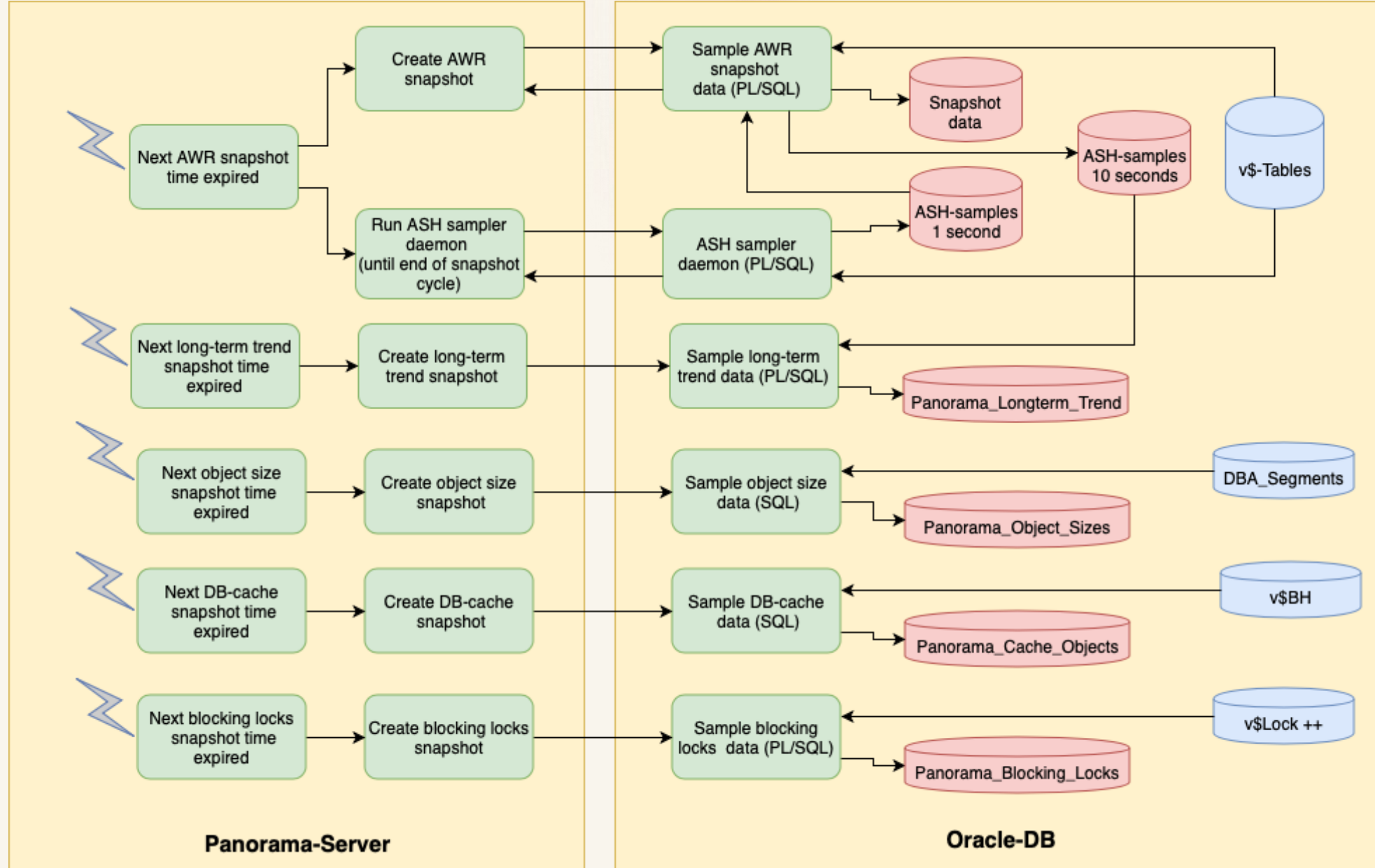
- All AWR views that are used in Panorama's monitoring and analysis functions will have a similar replacement in Panorama-Sampler
- Currently these are:

gv\$Active_Session_History
DBA_Hist_Active_Sess_History
DBA_Hist_Cache_Advice
DBA_Hist_Database_Instance
DBA_Hist_Datafile
DBA_Hist_Enqueue_Stat
DBA_Hist_FileStatXS
DBA_Hist_IOStat_Detail
DBA_Hist_IOStat_Filetype
DBA_Hist_Log
DBA_Hist_Memory_Resize_Ops
DBA_Hist_OSStat
DBA_Hist_OSStat_Name

DBA_Hist_Parameter
DBA_Hist_PGASat
DBA_Hist_Process_Mem_Summary
DBA_Hist_Resource_Limit
DBA_Hist_Seg_Stat
DBA_Hist_Service_Name
DBA_Hist_Service_Stat
DBA_Hist_SGASat
DBA_Hist_Snapshot
DBA_Hist_SQL_Bind
DBA_Hist_SQL_Plan
DBA_Hist_SQLStat
DBA_Hist_SQLText

DBA_Hist_StatName
DBA_Hist_Sysmetric_History
DBA_Hist_Sysmetric_Summary
DBA_Hist_System_Event
DBA_Hist_SysStat
DBA_Hist_Tablespace
DBA_Hist_Tempfile
DBA_Hist_TempStatXS
DBA_Hist_TopLevelCall_Name
DBA_Hist_UndoStat
DBA_Hist_WR_Control

Technical implementation of the Panorama-Sampler



Limits of technical implementation

Active Session History (ASH) has some restrictions compared to the original

- The SQL plan ID and operation are not recorded due to a lack of source. Unfortunately, this means that the load cannot be broken down by ASH into the individual lines of the execution plan.

- But there is an alternative for EE without Diagnostics Pack:

```
statistics_level=ALL or /*+ GATHER_PLAN_STATISTICS */
```

The `gv$sql_plan_statistics_all.elapsed_time` is per plan line

- Only top-level SQL statements are recorded as listed in `v$Session.SQL_ID`. This means that recursively executed SQL statements are not reported.
- I/O requests and amount read/written are not recorded because sampling from `v$sesstat` is too slow for one sample record per second and session.

Configuration of Panorama-Sampler

- Config attribute PANORAMA_MASTER_PASSWORD at Panorama start controls:
 - Login as admin enables additional control and configuration functions
 - Sampling will be executed on configured target DBs
- Configured data is stored encrypted in the local file system of the Panorama instance.
- Important: Set storage location by config setting PANORAMA_VAR_HOME, otherwise the configuration of target DBs will only be temporary.

Storage

I/O analysis

SGA/PGA-Details

Spec. additions

Admin

Help

Enter master password for access

Master-password

Dragnet investigation

Execute with given parameters

SQL worksheet

Admin login

SGA/PGA-Details

Spec. additions

Admin

Help

Configuration (stored in local file system)

Last AWR snapshot duration

Last object size snapshot time

Last object size snapshot duration

Last object size snapshot time

Panorama-Sampler config

Configure target databases for I

DB connection pool

Usage history

Configuration: DB connection

- Enter connection parameter and user credentials for DB
- Optionally define a different schema for data objects (mandatory if user = SYSTEM)
- Config data is encrypted with server key and PANORAMA_MASTER_PASSWORD

Add new Panorama-Sampler configuration

Database connect info

AWR and ASH

Object size history

DB-cache history

Blocking locks history

Long-term trend

ID	1
Config name	
DB identification mode	☒ TNS-Names ☐ Host, Port, SID/Service-Name
TNS-Alias	DOCKER_122_EE_PDB1 - 10.211.55.2 : 1521 : ORCLPDB1 Filter
Username	
Password	
Schema owner	

Test connection

Save

Cancel

Configuration: AWR and ASH counterparts

- Snapshot cycle and retention time are similar to AWR
- The limits for minimum number of executions and minimum runtime within a snapshot cycle allow reduction to relevant SQL for recording.
SQLs with negligible runtime that are executed infrequently can be excluded.
- The 1-second ASH samples can also be retained beyond the AWR snapshot, which reduces them into 10-second samples.

Add new Panorama-Sampler configuration

Database connect info

AWR and ASH

Object size history

DB-cache history

Blocking locks history

Long-term trend

Sample AWR and ASH	☐	
Snapshot cycle (minutes)	60	
Snapshot retention (days)	32	
Limits for SQL-history	Min. executions 2	Min. runtime (ms) 10
Keep time for 1-second ASH samples (hours)	3	

Test connection

Save

Cancel

Using Panorama-Sampler with RAC and PDB

Rules for Real Application Clusters (RAC):

- Panorama Sampler records for the RAC instance of the configured connection only.
- Therefore, used TNS services should be permanently assigned to a specific RAC node.
- A separate configuration must be created in Panorama Sampler for each RAC node.
- The data for the various RAC nodes should be stored in the same schema (and tables).

Rules for Pluggable Database (PDB):

- The SYSTEM user in CDB can sample the data for CDB and all PDBs at once
- Non-system users in CDB only sample the data of the CDB.
- In this case, a separate configuration is required for each PDB.

Redirect access on AWR views to Panorama replacements

Other analysis software can also be used without license violation by redirecting the access on AWR/ASH views to the Panorama replacement by synonyms.

```
SET SERVEROUTPUT ON;
BEGIN
  FOR Rec IN (SELECT v.View_Name, p.Object_Name Panorama_Name
                FROM   DBA_VIEWS v
                LEFT OUTER JOIN (SELECT /*+ NO_MERGE */ Object_Name
                                FROM   User_Objects
                                WHERE  Object_Name LIKE 'PANORAMA_%'
                                ) p ON p.Object_Name = REPLACE(v.View_Name, 'DBA_HIST_', 'PANORAMA_')
                WHERE  v.View_Name LIKE 'DBA_HIST_%' AND v.Owner = 'SYS'
                ORDER BY Panorama_Name NULLS FIRST, v.View_Name
              )
  LOOP
    IF Rec.Panorama_Name IS NULL THEN
      DBMS_OUTPUT.PUT_LINE('Not substituted by Panorama, redirecting synonym to not existing object: '||Rec.View_Name);
      EXECUTE IMMEDIATE 'CREATE OR REPLACE SYNONYM '||Rec.View_Name||' FOR NOT_SUBSTITUTED';
    ELSE
      DBMS_OUTPUT.PUT_LINE('Substituted by Panorama, redirecting synonym '||Rec.View_Name||' to '||Rec.Panorama_Name);
      EXECUTE IMMEDIATE 'CREATE OR REPLACE SYNONYM '||Rec.View_Name||' FOR '||Rec.Panorama_Name;
    END IF;
  END LOOP;
  DBMS_OUTPUT.PUT_LINE('Substituted by Panorama, redirecting synonym GV$ACTIVE_SESSION_HISTORY to PANORAMA_V$ACTIVE_SESS_HISTORY');
  EXECUTE IMMEDIATE 'CREATE OR REPLACE SYNONYM GV$ACTIVE_SESSION_HISTORY FOR PANORAMA_V$ACTIVE_SESS_HISTORY';
  DBMS_OUTPUT.PUT_LINE('Substituted by Panorama as view V$ACTIVE_SESSION_HISTORY on PANORAMA_V$ACTIVE_SESS_HISTORY');
  EXECUTE IMMEDIATE 'CREATE OR REPLACE VIEW V$ACTIVE_SESSION_HISTORY AS
                    SELECT * FROM PANORAMA_V$ACTIVE_SESS_HISTORY
                    WHERE Inst_ID = SYS_CONTEXT(''USERENV'', ''INSTANCE'')';
END;
/
```

Demo

- Start of Panorama instance
- Configuration of DB target for Panorama-Sampler
- Panorama functions based on sampled data
 - Dashboard based on 1-second ASH records
 - Session wait historic based on whole ASH records (1 and 10-second samples)
 - SQL history, SQL details for certain period, summary of snapshots for SQL
 - Historic execution plans
 - Object size evolution
 - History of DB-cache usage
 - Long-term trend evolution

Thank you for your interest



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